

RF-Robust Payload-Invariant Noise-Masked Low Probability of Intercept Waveform Synthesis with Authenticated Encryption for Software Defined Radio Links

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Abstract—Low Probability of Intercept (LPI) transmission reduces detectability at an intercept receiver (warden) while maintaining reliable communication. With machine learning (ML)-based electronic intelligence (ELINT) receivers, power spectral density (PSD) camouflage alone is insufficient, as detectors exploit time-frequency structure and higher-order statistics. This paper presents an RF-robust, payload-invariant (RR-PI) noise-masked waveform synthesis and authenticated payload stack for practical software-defined radio (SDR) links. The transmitter generates complex baseband I/Q sequences with marginal PSD statistics close to an AWGN reference (KS-PSD 0.022), while payload integrity is ensured using Reed–Solomon coding and AES-GCM authenticated encryption. We evaluate pre-FEC BER, KS-PSD, and peak-to-average power ratio (PAPR) across signal-to-noise ratio (SNR) and embedding blend factor, characterizing the stealth–reliability trade-off. Detectability is assessed using PSD- and time-frequency-domain CNN wardens under an AWGN null hypothesis. Over-the-air bladeRF experiments confirm feasibility and identify synchronization and RF impairment compensation as key bottlenecks for authenticated payload recovery.

Index Terms—Low probability of intercept (LPI), software defined radio (SDR), machine learning, RF signal detectability, false alarm rate (FAR), power spectral density (PSD), time-frequency analysis (STFT), Reed–Solomon coding, authenticated encryption (AES-GCM), payload invariance

I. INTRODUCTION

Low Probability of Intercept (LPI) communication aims to reduce the probability that an intercept receiver (warden) detects or classifies a transmission while maintaining useful information transfer. Covert communication theory establishes limits on covert throughput under statistical detection constraints [1]. In modern ELINT, DNN/CNN models perform modulation recognition and signal classification using raw I/Q, PSD, and time–frequency (TF) representations [2], [3], motivating transmitter-centric waveform shaping that suppresses stable spectral and TF signatures.

Adversarial machine learning (AML) studies further show that detector outcomes depend on channel constraints and over-the-air effects [4]–[7]. Many masking works remain simulation-centric and report limited system-level validation. This paper therefore emphasizes an SDR-realizable pipeline and a reproducible protocol reporting both authenticated

communication behavior and ML warden outcomes under fixed preprocessing. The main contributions are:

- 1) An RR-PI SDR pipeline integrating AES-GCM, RS coding, pilot-aided synchronization, and noise-masked I/Q synthesis.
- 2) A sweep over SNR and ε reporting BER, KS-PSD, and PAPR.
- 3) A balanced dataset and two wardens (PSD-CNN, TF-CNN) showing representation-dependent detectability.
- 4) Hardware evidence using bladeRF captures.

Fig. 1 summarizes the end-to-end RR-PI SDR pipeline, highlighting the cryptographic framing, burst synthesis, SDR interface, and receiver synchronization stages.

II. SYSTEM MODEL AND THREAT MODEL

A. RF Baseband Model

Let $x[n] \in \mathbb{C}$ denote transmitted I/Q at sampling rate f_s . The received samples are modeled as

$$y[n] = (h[n] * x[n]) e^{j(2\pi\Delta f n / f_s + \phi[n])} + \eta[n], \quad (1)$$

where $h[n]$ models channel response, Δf is carrier frequency offset (CFO), $\phi[n]$ captures phase drift, and $\eta[n]$ denotes noise.

To reduce detectability in the *PSD domain*, the burst is power-normalized and formed as a controlled blend between a noise draw $n_0[n] \sim \mathcal{CN}(0, \sigma^2)$ and an embedding component $s[n]$ (projection or learned decoder-encoder). A representative form is

$$x[n] = \alpha(\sqrt{1 - \varepsilon} n_0[n] + \sqrt{\varepsilon} s[n]), \quad (2)$$

where α enforces target power and ε controls the stealth–reliability trade-off, explaining why KS-PSD can remain near the AWGN reference while bit recovery varies with ε .

B. Warden Models

Two wardens are considered: (i) a PSD-domain CNN operating on Welch PSD vectors and (ii) a TF-domain CNN operating on STFT spectrograms. The warden is passive, has no secret keys, and trains supervised models on labeled examples under fixed preprocessing.

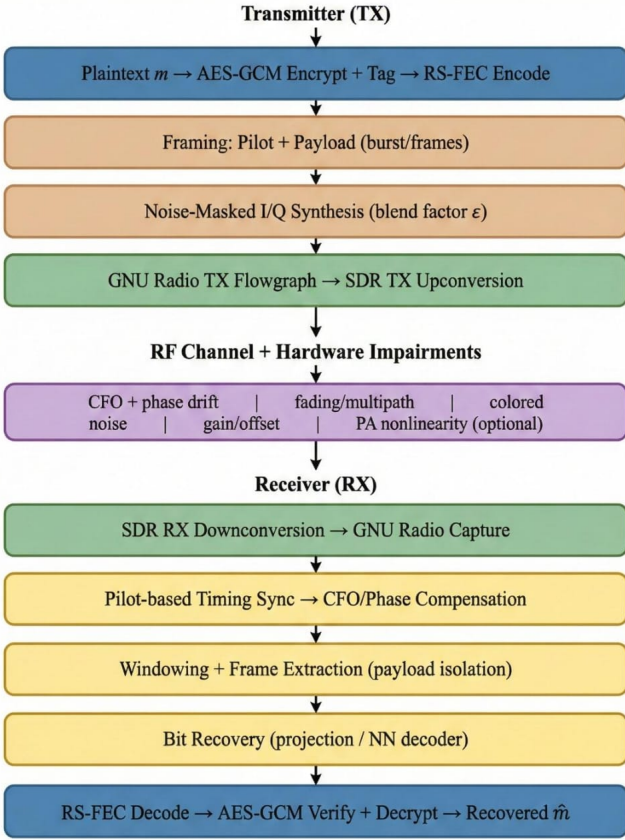


Fig. 1. End-to-end RR-PI SDR pipeline with cryptographic framing, SDR interface, and synchronization processing stages. Color coding: encryption/FEC (blue), frame synthesis (tan), SDR operations (green), synchronization/compensation (yellow), RF channel effects (purple).

III. SDR IMPLEMENTATION AND DATASET

A. Hardware Configuration

The SDR (bladeRF) is configured at center frequency 433.5 MHz with sampling rate $f_s = 2$ M sample per second. Two antennas are separated by 0.3 m indoors (line-of-sight) using the internal reference clock; transmit/receive gains are 17 dB/20 dB.

B. Burst Structure and Payload Stack

Each burst contains a 1024-sample Zadoff–Chu pilot and a 1536-sample payload (six frames of 256 samples), totaling 2560 samples. The plaintext is protected by AES-GCM (confidentiality + integrity). RS coding improves tolerance to residual bit errors prior to authentication; authenticated decode succeeds only if synchronization and impairments are corrected sufficiently to meet RS correction and preserve the AES-GCM tag.

C. Receiver Processing: Timing, CFO, and Phase

The receiver searches for the pilot using correlation between the known pilot $p[n]$ and incoming samples $r[n]$:

$$\hat{\tau} = \arg \max_{\tau} \left| \sum_{n=0}^{N_p-1} r[n + \tau] p^*[n] \right|. \quad (3)$$

After coarse timing, CFO is estimated from the pilot correlation phase progression, and a constant CFO $\hat{\Delta f}$ is applied:

$$\tilde{r}[n] = r[n] \exp(-j2\pi\hat{\Delta f} n/f_s). \quad (4)$$

A constant phase correction follows using the pilot reference. Hardware results indicate residual CFO/phase drift and framing errors dominate authenticated recovery even when PSD-level camouflage is preserved.

D. SDR Evidence Plots (from bladeRF captures)

Fig. 2 shows the bladeRF TX/RX testbed and GNU Radio flowgraphs used for over-the-air capture. Figs. 3–4 provide representative TX/RX magnitude and spectrum evidence from captured bursts, and Fig. 5 shows the corresponding QT GUI visualization during live SDR operation.

E. Detectability Dataset

The dataset contains 1200 I/Q examples of length 1536 with 4 balanced classes: AWGN, QPSK, OFDM, and RR-PI (300 each). A stratified split (seed 123) yields train/validation/test sizes of 720/240/240 with per-class counts 180/60/60. For reproducibility, all classes are power-normalized: AWGN is circularly symmetric complex Gaussian noise, QPSK is pulse-shaped, OFDM uses IFFT with cyclic prefix, and RR-PI is generated by the proposed pipeline. All bursts are cropped to the same 1536-sample payload window used for warden training.

IV. FEATURE EXTRACTION AND ML DETECTORS

A. PSD Features (Welch)

Welch PSD uses a Hann window with $n_{\text{perseg}} = 1024$, $n_{\text{overlap}} = 512$, and $n_{\text{fft}} = 2048$ at $f_s = 2$ MHz [8]. The two-sided spectrum is FFT-shifted, converted to dB, and z-score normalized per example (2048-D feature). KS-PSD is computed as $\text{KS} = \sup_z |F_{\text{burst}}(z) - F_{\text{AWGN}}(z)|$ under matched Welch settings.

B. TF Features (STFT)

STFT uses a Hann window with $n_{\text{perseg}} = 256$, $n_{\text{overlap}} = 192$, and $n_{\text{fft}} = 512$ [9]. The 512×21 magnitude spectrogram is FFT-shifted, converted to dB, and normalized per example; this retains short-time structure that may remain detectable even under PSD camouflage.

C. Detector Training

Both wardens are trained with Adam (learning rate 10^{-3}) for 20 epochs using cross-entropy loss; accuracy and macro-F1 are reported on the held-out test split. Preprocessing (z-score) is applied consistently across all classes.

V. RESULTS

A. Stealth–Reliability Trade-off (CSV-derived sweep)

A sweep over $\text{SNR} \in \{10, 15, 20, 25\}$ dB and $\varepsilon \in \{0.10, 0.12, 0.14\}$ is performed. Fig. 6 shows BER decreases with larger ε , Fig. 7 reports the corresponding KS-PSD behavior (remaining near ≈ 0.022), and Fig. 8 shows the associated PAPR trends across the same sweep.

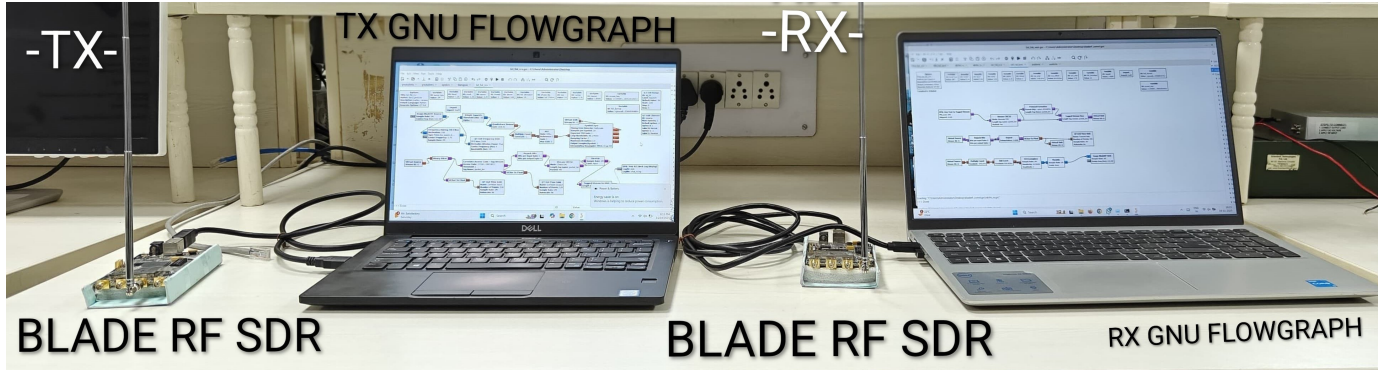


Fig. 2. bladeRF over-the-air SDR testbed showing TX/RX GNU Radio flowgraphs and the corresponding SDR units used for burst transmission and capture.

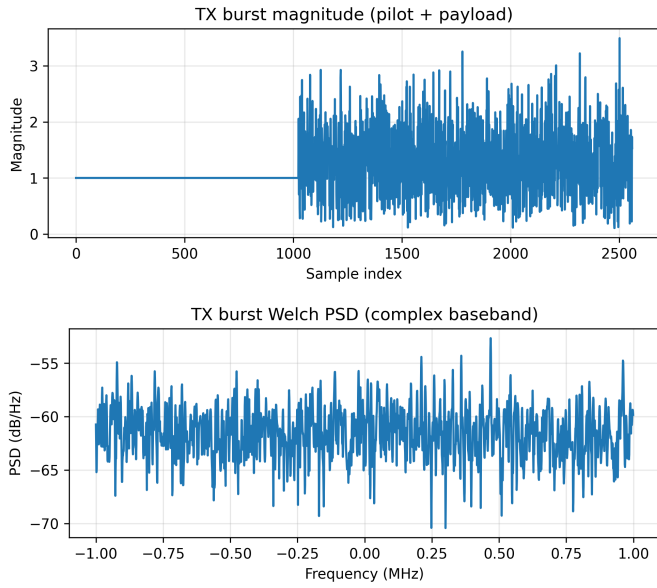


Fig. 3. Over-the-air SDR evidence from bladeRF captures (TX burst magnitude and spectrum).

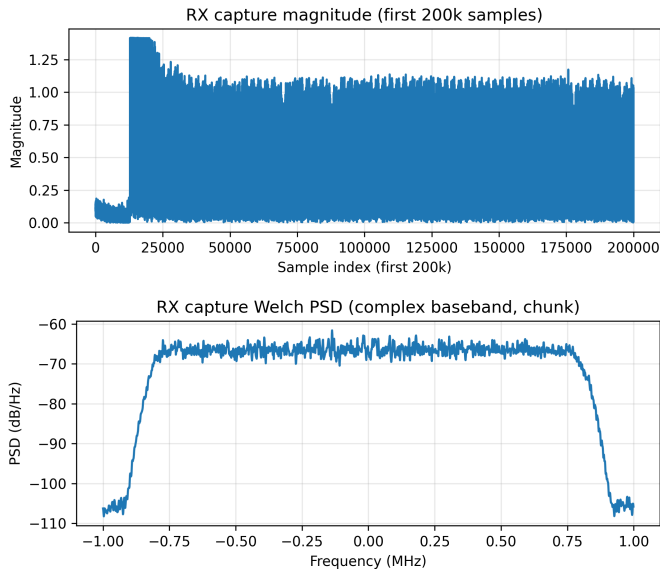


Fig. 4. Over-the-air SDR evidence from bladeRF captures (RX capture magnitude and spectrum).

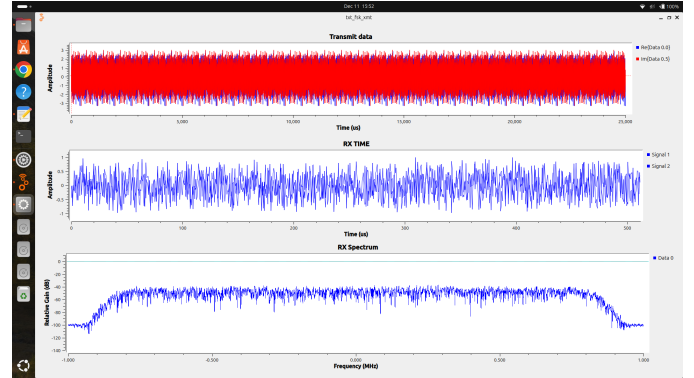


Fig. 5. GNU Radio QT GUI visualization of transmitted/received baseband and spectrum during over-the-air SDR operation.

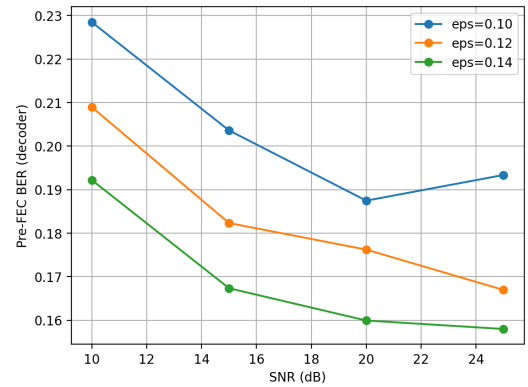


Fig. 6. Metric sweep across SNR and ε : BER (from evaluation CSV logs).

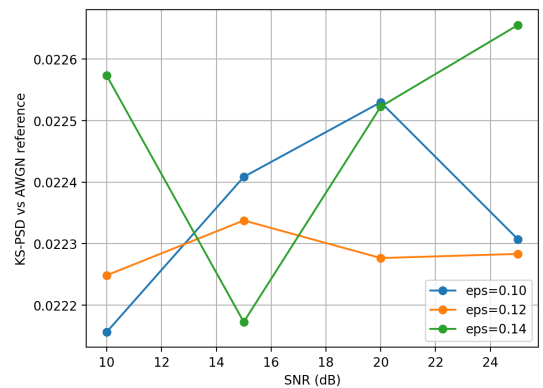


Fig. 7. Metric sweep across SNR and ε : KS-PSD (from evaluation CSV logs).

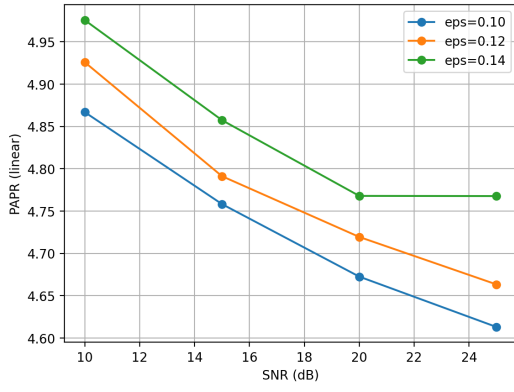


Fig. 8. Metric sweep across SNR and ε : PAPR (linear ratio) (from evaluation CSV logs).

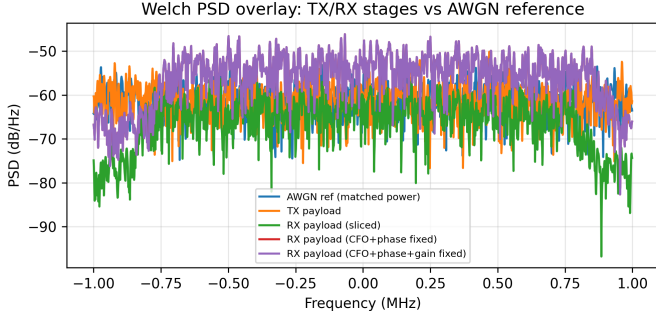


Fig. 9. Welch PSD overlay: AWGN reference vs. RR-PI bursts (same preprocessing as KS-PSD).

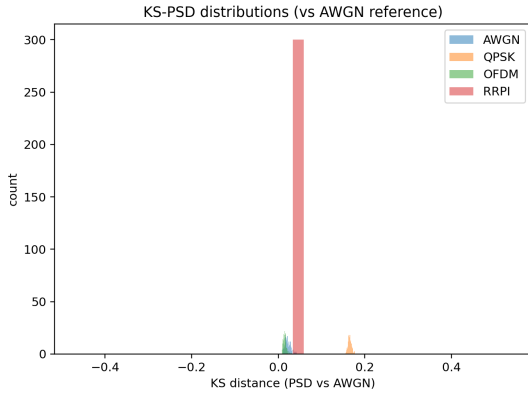


Fig. 10. KS-PSD histogram versus an AWGN reference under matched PSD settings.

B. KS-PSD Distribution Shape

Fig. 10 shows the KS-PSD histogram versus an AWGN reference under matched PSD settings; the clustering near low KS indicates the synthesized bursts largely track the intended PSD distribution across evaluated conditions. The PSD overlay in Fig. 9 visually supports this behavior by showing close agreement between the AWGN reference and RR-PI bursts under the same preprocessing.

C. ML Warden Outcomes

Figs. 11–14 report warden outcomes under fixed preprocessing. PSD-CNN attains higher accuracy, consistent with small systematic PSD differences remaining separable. TF-CNN shows increased confusion between AWGN and structured

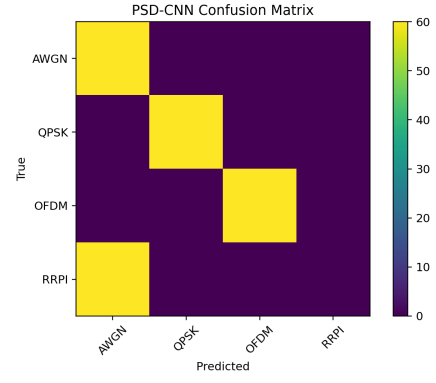


Fig. 11. PSD-CNN confusion matrix on the test set.

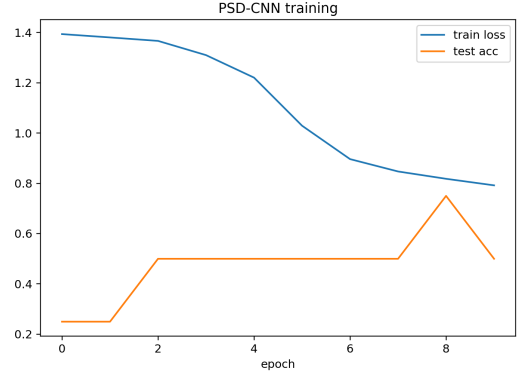


Fig. 12. PSD-CNN training/validation accuracy.

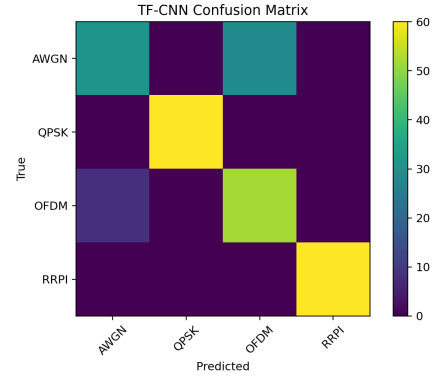


Fig. 13. TF-CNN confusion matrix on the test set.

modulations (notably OFDM), while RR-PI is retained as a distinct class, motivating FAR/TPR reporting under an AWGN null hypothesis.

a) Interpretation of Multi-Class Accuracy vs. RR-PI Detectability: High multi-class accuracy can be dominated by separating structured modulations (QPSK/OFDM) from noise-like classes; confusion-matrix FAR/TPR under an AWGN null hypothesis therefore provides an operational LPI view.

D. Domain-Specific Separability and Error Patterns

The warden results indicate representation-dependent behavior: low KS-PSD implies close *marginal* spectral power statistics under Welch averaging, yet discriminative models can exploit higher-order structure (e.g., slight spectral tilt, windowing/framing artifacts, or short-time structure). Small decision

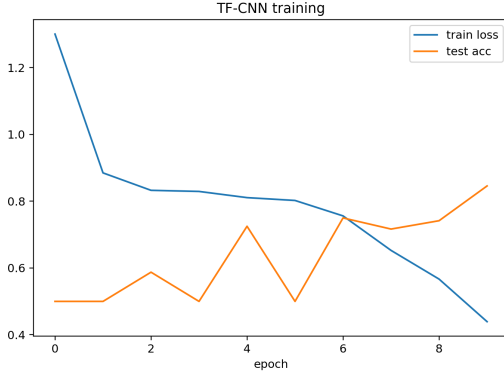


Fig. 14. TF-CNN training/validation accuracy.

margins also imply mild domain shifts (CFO/gain/colored noise) can change predictions, aligning with hardware observations and motivating TF-aware constraints and robustness evaluation under controlled domain shifts.

Representation-Dependent Stealth: Why Low KS-PSD Does Not Imply ML Indistinguishability: KS-PSD compares marginal distributions of spectral power values and is insensitive to temporal correlation, phase structure, and framing artifacts. CNN wardens can exploit any stable divergence induced by practical synthesis/processing (e.g., band-limiting, windowing edges, slight spectral tilt, or persistent short-time structure). Thus, low KS-PSD is necessary but not sufficient for stealth, motivating multi-representation evaluation beyond a single metric.

E. Interpretation of FAR/TPR Under an AWGN Null Hypothesis

Beyond accuracy, an operational LPI criterion is the false alarm rate (FAR) under a noise-only null hypothesis: how often a warden declares RR-PI (or “any signal”) when the input is pure AWGN. For confusion-matrix counts $C(i \rightarrow j)$ (true i , predicted j) and N_i test examples of class i , we define

$$\text{FAR}_{\text{RRPI} \leftarrow \text{AWGN}} = \frac{C(\text{AWGN} \rightarrow \text{RRPI})}{N_{\text{AWGN}}}, \quad (5)$$

$$\text{FAR}_{\text{AnySig} \leftarrow \text{AWGN}} = \frac{N_{\text{AWGN}} - C(\text{AWGN} \rightarrow \text{AWGN})}{N_{\text{AWGN}}}.$$

The RR-PI true positive rate (TPR) is

$$\text{TPR}_{\text{RRPI}} = \frac{C(\text{RRPI} \rightarrow \text{RRPI})}{N_{\text{RRPI}}}. \quad (6)$$

From Figs. 11 and 13, $N_{\text{AWGN}} = N_{\text{RRPI}} = 60$ per class.

a) PSD-CNN (Welch PSD): Fig. 11 yields $\text{FAR}_{\text{AnySig} \leftarrow \text{AWGN}} = 0.0\%$ and $\text{FAR}_{\text{RRPI} \leftarrow \text{AWGN}} = 0.0\%$, while $\text{TPR}_{\text{RRPI}} = 0.0\%$ because RR-PI is predicted as AWGN. Under Welch-PSD features, RR-PI collapses toward AWGN, consistent with deliberate PSD-domain camouflage.

b) TF-CNN (STFT): Fig. 13 yields $\text{FAR}_{\text{RRPI} \leftarrow \text{AWGN}} = 0.0\%$ and $\text{FAR}_{\text{AnySig} \leftarrow \text{AWGN}} = 50.0\%$ (AWGN is often predicted as a structured class, primarily OFDM), while $\text{TPR}_{\text{RRPI}} = 100.0\%$.

TABLE I

FAR ON AWGN AND RR-PI DETECTION RATE (TPR) INFERRED FROM THE SHOWN CONFUSION MATRICES (FIGS. 11, 13) USING $N_{\text{AWGN}} = N_{\text{RRPI}} = 60$ AND (5)–(6).

Warden	$\text{FAR}_{\text{RRPI} \leftarrow \text{AWGN}}$	$\text{FAR}_{\text{AnySig} \leftarrow \text{AWGN}}$	TPR_{RRPI}
PSD-CNN	$0/60 = 0.0\%$	$0/60 = 0.0\%$	$0/60 = 0.0\%$
TF-CNN	$0/60 = 0.0\%$	$30/60 = 50.0\%$	$60/60 = 100.0\%$

TABLE II

MEANS ACROSS SNR POINTS FROM THE METRIC SWEEP (LOWER BER AND KS-PSD ARE BETTER; LOWER PAPR IS PREFERRED). PAPR IS A linear ratio ($\max |x[n]|^2 / \mathbb{E}[|x[n]|^2]$).

ϵ	BER_{dec}	KS-PSD	PAPR	BER_{proj}
0.10	0.2032	0.02235	4.7276	0.0057
0.12	0.1836	0.02229	4.7749	0.0035
0.14	0.1694	0.02248	4.8421	0.0022

TABLE III

ML-BASED DETECTABILITY (TEST SET).

Detector	Accuracy (%)	Macro-F1
PSD-CNN	93.89	0.939
TF-CNN	87.22	0.863

c) Interpretation: These results reinforce representation dependence: PSD features obscure RR-PI separability (low FAR, low TPR), whereas TF features detect RR-PI strongly but show higher “any-signal” false alarms on AWGN. FAR/TPR therefore complements multi-class accuracy for an operational LPI view.

d) Binary vs. Multi-Class Interpretation: FAR/TPR under the AWGN null highlights deployment-relevant behavior that multi-class accuracy can mask: PSD-CNN yields zero AWGN false alarms but does not separate RR-PI from AWGN, whereas TF-CNN detects RR-PI strongly but has elevated “any-signal” false alarms on AWGN.

Note that aggregate accuracy can be dominated by separating structured modulations (QPSK/OFDM) from noise-like classes; class-specific FAR/TPR under an AWGN null hypothesis is captured in Table I.

F. Hardware Implications for Authenticated Recovery

Authenticated recovery is constrained by synchronization and impairment compensation. AES-GCM verification is a strict pass/fail gate: residual timing/CFO/phase errors that cause moderate bit corruption can defeat tag verification despite an otherwise noise-like spectrum, so receiver RF robustness is as critical as waveform stealth.

G. Receiver Error Sensitivity and Practical Design Implications

Hardware results suggest a synchronization-dominated failure mode: (i) small timing slips change payload alignment and can break RS codeword structure; (ii) residual CFO/phase causes rotation and time-varying phase that a fixed projection/decoder may not tolerate, increasing burst-to-burst error variance. Lightweight receiver upgrades (pilot re-detection, decision-directed phase tracking, and frame boundary checks) are the most direct path to improving authenticated recovery without weakening stealth.

TABLE IV
COMPARISON WITH REPRESENTATIVE RF MACHINE LEARNING AND
COVERT/LPI-RELATED LITERATURE.

Work	Goal / Setting	ML Role	Key Takeaway
Bash <i>et al.</i> [1]	Covert communication theory	Information theoretic	Theoretical detection bounds only
O'Shea <i>et al.</i> [2]	RF classification from I/Q	CNN feature learning	ML enables powerful RF wardens
West & O'Shea [3]	Deep modulation recognition	Deep CNNs	TF structure is learnable
Flowers <i>et al.</i> [4]	Adversarial RF evasion	AML evaluation	Hardware effects
Lin <i>et al.</i> [5]	Adversarial attacks	DNN-based classifiers	ML-aware threat model
Restuccia <i>et al.</i> [7]	Waveform manipulation	Adversarial learning	RF Channel-aware evaluation needed
This work	Noise-masked over SDR	LPI Neural synthesis + decode	KS-PSD + BER + PAPR + SDR evidence

H. Comparative Analysis Against Prior RF Machine Learning and Covert/LPI Literature

Table IV highlights that this work reports an SDR-oriented, reproducible evaluation spanning KS-PSD, BER, and PAPR together with dual-domain ML wardens and hardware evidence.

VI. LIMITATIONS AND FUTURE WORK

Authenticated recovery in the proposed system depends critically on accurate timing synchronization, carrier frequency offset (CFO) and phase compensation, and robust frame detection under multipath propagation and analog front-end impairments. While low KS-PSD effectively reduces separability in the PSD domain, TF-domain wardens can still exploit short-time structure and higher-order statistics that are not captured by marginal spectral metrics alone. Moreover, residual RF impairments that are tolerable in terms of average BER can remain catastrophic for authenticated payload recovery due to the strict pass/fail nature of AES-GCM verification.

Future work will therefore incorporate explicit time-frequency domain constraints during waveform synthesis, jointly optimizing stealth and robustness across multiple representations. In addition, robustness will be evaluated under controlled domain shifts, including gain variation, CFO jitter, mild filtering, and hardware-induced distortions, to better reflect deployment conditions. Further extensions will replicate a broader set of published warden architectures and baselines under a unified and consistent preprocessing pipeline, enabling fairer cross-work comparison and deeper insight into representation-dependent detectability.

VII. CONCLUSION

This paper evaluated an SDR-realizable RF-robust, payload-invariant (RR-PI) noise-masked LPI waveform synthesis and authenticated payload stack using communication, stealth, and detectability metrics. Performance was quantified using BER, KS-PSD, and PAPR, together with ML warden detectability in both PSD and TF domains. Across the evaluated SNR range and embedding blend factor ε , KS-PSD remains tightly bounded near ≈ 0.022 while BER improves monotonically with increasing ε , highlighting a clear stealth-reliability trade-off.

Over-the-air bladeRF experiments confirm practical feasibility and demonstrate that synchronization accuracy and RF impairment compensation dominate authenticated recovery performance, rather than spectral camouflage alone. ML-based results further show that detectability is strongly representation-dependent: high multi-class accuracy can be driven by structured modulations, while FAR/TPR under an AWGN null hypothesis exposes deployment-relevant LPI behavior. Collectively, these results emphasize that effective LPI evaluation must combine communication metrics, multi-representation ML analysis, and hardware validation to avoid overestimating stealth in realistic operational settings.

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